

1 | The decision

LUCKY CEMENT | MGT-353 | 5 October 2026

Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Lucky Cement's disclosed scale, not actual company records.

How many tonnes of Ordinary Portland Cement should Lucky Cement plan to supply to an illustrative Karachi/Sindh dealer territory over the next 12 weeks, and how much dispatch-linked production should it commit now versus keep flexible?

12 weeks: 5 October-21 December 2026. Owner: territory demand planner with production/dispatch managers. Planning unit: tonnes of Ordinary Portland Cement for an illustrative Karachi/Sindh dealer territory. The business risk is excessive prebuilding versus unmet orders.

Planning choice	Defined scope
Forecast object	Ordinary Portland Cement
Territory	Karachi / Sindh dealers
Operational horizon	12 weeks
Unit	Tonnes per week

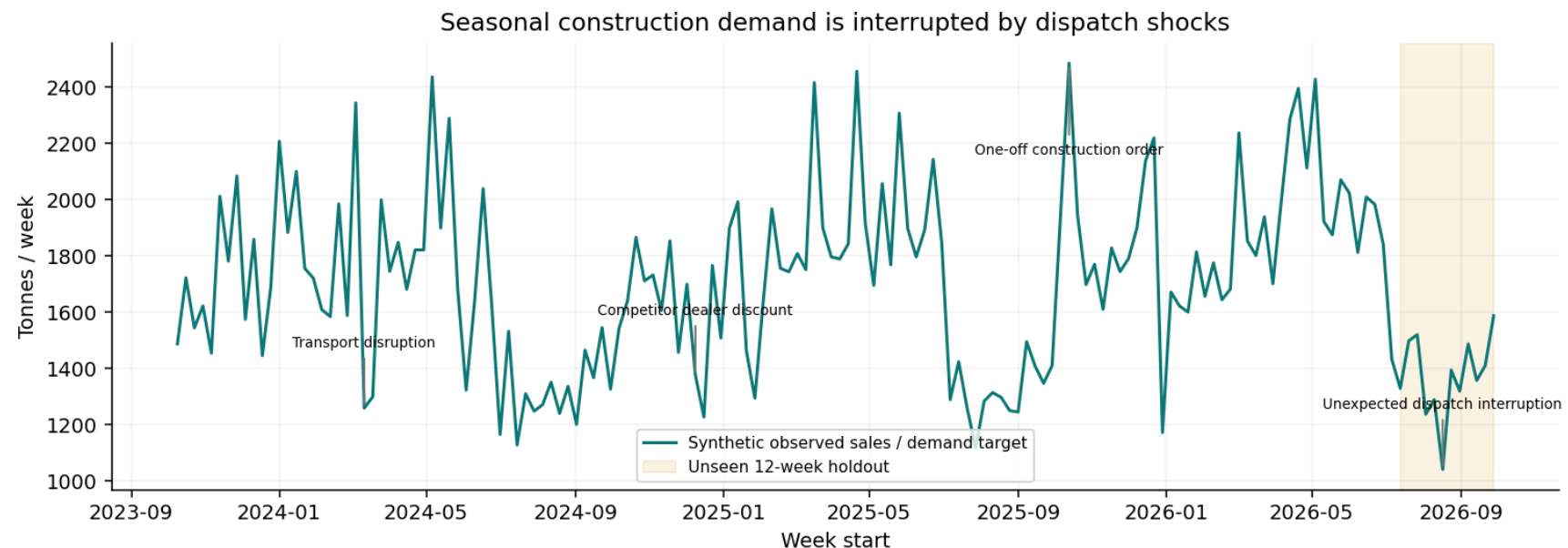
2 | Seasonal demand needs context

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Construction-season strength is interrupted by availability shocks.

The illustrative cement series combines a modest underlying growth trend with higher construction-season demand and a weaker monsoon block. Price changes occur in steps rather than every week. Dealer incentives create irregular dispatch lifts, while construction activity varies independently and contributes additional movement. Four exceptional incidents interrupt the normal pattern, including two availability-constrained episodes; these must not be interpreted as proof that customer demand disappeared. A simple level forecast may miss seasonal transitions, while a short moving average can be pulled down by recent stockouts. Seasonal naive forecasting provides a transparent benchmark but repeats last year's unusual weeks. A calendar seasonal-trend model can recognise month effects and gradual change, although it cannot anticipate unannounced disruptions. Model parameters and diagnosis for selection use training observations only; the full-history chart is descriptive, not permission to tune on holdout outcomes.



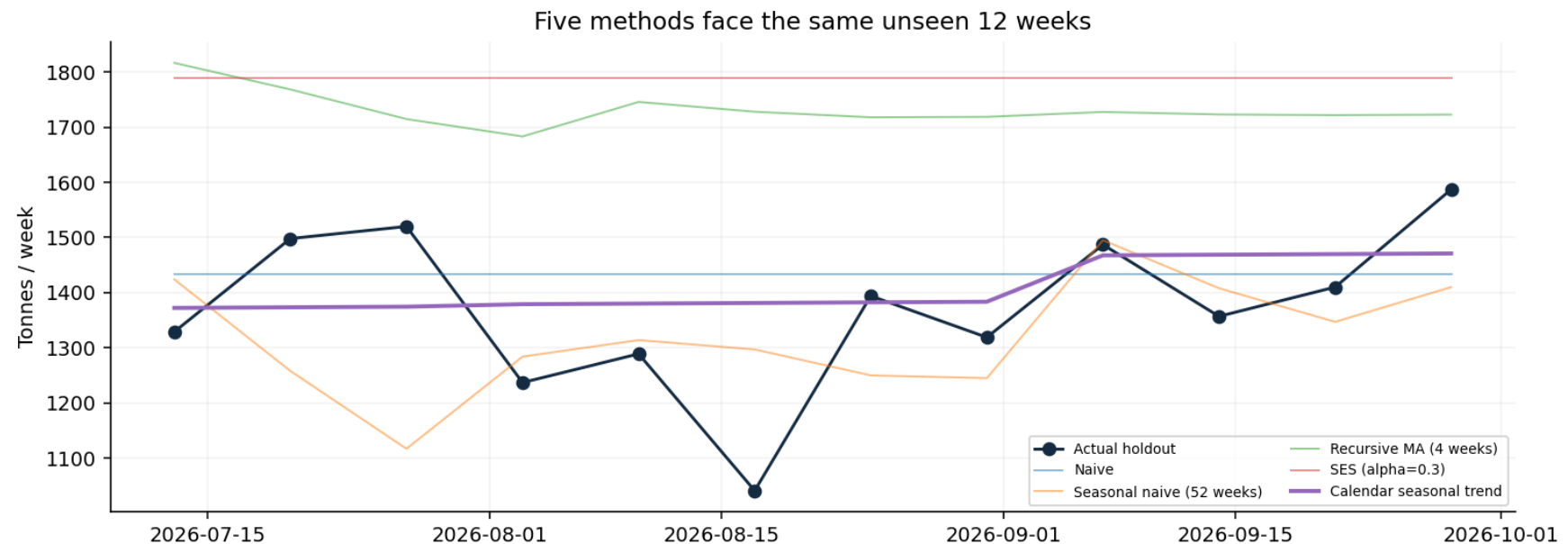
3 | A method earns provisional trust

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Calendar seasonal trend wins the common 12-week holdout.

WAPE 7.71% | MAPE 8.27% | MAD 106 t/week. Naive WAPE 8.79%. Five fixed methods use the same holdout; no actual holdout values feed recursive forecasts. The seasonal model is interpretable and uses known future dates, but this single comparison is not proof of stability.



4 | Drivers explain the synthetic story

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Dealer incentives and construction activity are associated with higher sales.

Training R-squared 0.857; adjusted R-squared 0.849. Promotion association +372 t/week; construction-season association +225 t/week; activity index +13.4 t per point. Price +PKR 1,000/t is associated with -47 t/week, but $p=0.275$: weak evidence. Time/price collinearity and dealer forward buying limit interpretation. These are built-in synthetic associations, not real Lucky Cement causal estimates.

Driver change	Associated weekly sales change	Evidence caveat
Dealer incentive active	+372 tonnes	May include dealer forward buying
Construction activity +1 point	+13.4 tonnes	Synthetic activity proxy
Price +PKR 1,000 per tonne	-47 tonnes	$p=0.275$; weak evidence

5 | Accuracy does not eliminate risk

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Tracking Signal -4.12: review the bias now.

Largest error: 17 August stockout week, overforecast by 340 tonnes. A 20% demand downside creates 4,422 tonnes excess under a fixed base plan; a two-week 40% dispatch loss still leaves 661 tonnes unserved after buffer. The +/-2 MAD band is illustrative; no 95% coverage claim is made.

Risk	Operational exposure	Managerial response
Demand 20% below base	4,422 tonnes excess under fixed plan	Reduce flexible releases
Two-week capacity disruption	661 tonnes unserved after buffer	Secure dispatch alternatives
Current bias signal	Tracking Signal -4.12	Review before plan release

6 | Commit a schedule, retain options

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Provisional schedule: 17,687 tonnes; keep 4,422 tonnes flexible.

Forecast total 22,109 tonnes, separate 212-tonne buffer. Reserve access to 20% upside capacity and release weekly. Complete the current stockout/bias review before operational release. Reforecast for >10% early-week deviation, Tracking Signal outside ± 4 or announced dispatch disruption. Validate the assumed 2,400 t/week capacity and one-week response before using this plan.

